

# Dual-task interference while receiving information on a head mounted display and manual tracking with and without auditory warnings<sup>☆</sup>

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## ABSTRACT

**Background:** Dual-task interference is a concern when users attempt to use head mounted displays (HMD) while performing seated manual control tasks. Auditory warnings of information displayed on the HMD may or may not assist users while multi-tasking.

**Objective:** In this study, we investigated the dual-task interference of words, to be remembered later, presented on a HMD and motor control in a manual tracking task. The word presentations were warned or not warned with auditory signals and the impact of audio warnings were examined.

**Methods:** Participants were required to perform five tasks: (1) a compensatory tracking task performed alone, (2) a word memory task performed alone, (3) a tracking and word memory task, (4) a word memory task with words preceded by aural signals, and (5) a tracking and word memory task with words preceded by aural signals. In addition, in the dual-tasks, half of the word presentations were paired with directions changes in the tracking task to test for immediate perceptual interference.

**Results:** There were significant dual task costs for both mean tracking error and later word recall. Additionally, participants took significantly longer to respond to motion changes paired with word presentations than motion changes not paired with word presentations. However, the impact of auditory warnings on tracking performance was nuanced.

**Conclusion:** Even with an in field of view transparent HMD momentary and sustained cognitive dual-task interference remains. Reaction times are affected most in the worst case scenario, when task critical events occur at the same time as a text message.

**Application:** The use of HMDs in time critical manual control tasks (such as operating machinery) should be limited. The use of aural warnings to alert operators to information displayed on an HMD requires further research; it may disrupt appropriate or natural task ordering.

Millions of head-mounted wearable computers and head mounted displays (HMDs) are expected to be sold in the coming years (Wickens et al., 2021; Levy, 2012). HMDs have many advantages over standard smart phones and other commonly used communication equipment. HMDs often use a transparent screen positioned within the wearer's field of vision to allow the wearer to attend to the screen with a simple saccade of the eye whereas a smart phone draws attention away from and occludes the real world. HMDs are designed to be always on and accessible while the user performs everyday tasks (walking, working, driving, climbing and other tasks). However, a poorly designed wearable experience may place new demands on limited human attention

capacity and even pose a danger by distracting from critical tasks. Thus there is a need to research how using a HMD may affect a user's ability to perform while doing other tasks.

Performing communication tasks while performing other tasks, such as driving or walking, can cause dual-task interference. Extensive previous research has demonstrated operating mobile phones causes users to significantly slow their walking pace (Lin et al., 2007; Lopresti-Goodman et al., 2012; Oulasvirta et al., 2005). Using mobile phones also causes the user to have reduced awareness of their surroundings (Hyman et al., 2010; Nasar et al., 2008). While using mobile phones, users often do not notice even highly unusual and salient events (Hyman et al.,

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2010). Users' spatial reasoning is also impaired. For example, users have a higher rate of misjudging the speed of oncoming objects (Schwebel et al., 2012) or whether they would be able to fit through gap without turning (Lopresti-Goodman et al., 2012). The number of injuries involving distraction by mobile phones has been increasing year to year (Nasar and Troyer, 2013; Smith et al., 2013). Receiving calls, reading text messages and, in particular, sending texts messages of their own have all been shown to be associated with significantly increased rates of vehicle crashes resulting in injury or death (Klauer et al., 2014; McEvoy et al., 2005; Strayer et al., 2006). A study of 48 novice drivers found that while sending or receiving texts the drivers had nearly a four times greater likelihood of being in a crash or near crash compared with when they were undistracted (Klauer et al., 2014); a relative increase in risk similar to driving with a blood alcohol level between 0.5 and 0.8 g/L (Hels et al., 2013). Researchers have shown that hands-free phones are a significant hazard when used while driving (Collet et al., 2010; Ishigami and Klein, 2009; McEvoy et al., 2005; Pattern et al., 2004; Strayer et al., 2006).

In regards to HMDs specifically, Sawyer et al. (2014) compared the effects of an HMD and smartphone use on reaction times and braking recovery while driving a simulated vehicle. They found that, although participants recovered from brake events faster while using the HMD compared with the smartphone, reaction times to the event itself did not differ between devices. A number of additional studies have demonstrated some benefits for HMDs over head-down displays, but the use of HMDs still has interference effects on paired task performance (Wickens et al., 2021). In many of these studies the paired task, as in the case of Sawyer et al. (2014), was a driving task (He et al., 2015, 2018; Tippey et al., 2017; Young et al., 2016). In He et al. (2018) both the text messages presented on the HMD (Google Glass) and text messages presented on a head-down display were audial alerted. The usefulness of audial warnings for HMDs warrants further investigation. Audial warnings may or may not be particularly useful for HMDs as the visually displayed information is already within the user's visual field of view.

Similar to Sawyer et al.'s (2014) findings with driving, Woodham et al. (2016) examined the use of an HMD on the ability of climbers to traverse a climbing wall. Woodham and colleagues had participants climb a traverse without another task (single-task), climb while words were presented on a HMD (to be recalled later) with both an auditory warning or no warning preceding word presentation condition, and also had participants perform the word recall tasks while seated with no other demands (single-tasks). They found significant dual-task interference with both climbing and later word recall impaired by dual-tasking. However, they found no beneficial effects for auditory warning signals.

HMDs open up a large number of dual task scenarios as they can be used while people are on the move; driving (He et al., 2015, 2018; Tippey et al., 2017; Sawyer et al., 2014; Young et al., 2016) as demonstrated by Woodham et al.'s (2016) research on climbing, will not be the only task impacted. In the present study we performed an experiment similar to Woodham et al.'s utilizing five task conditions, except instead of climbing, we employed a seated manual tracking task. For the tracking task, participants were required to maintain the position of a circular target subject to a shifting drag force. This compensatory tracking was used as an analogue for other seated manual position control tasks (for example, flying a plane, steering a boat, or controlling a loader arm on a construction or agricultural machine) as it requires a serial detection, reaction, and stabilization response when the drag force changes. The tracking task also requires sustained attention to maintain position between drag force changes. Our choice in pairing the word presentation with later recall task utilized by Woodham et al. (2016) with this manual tracking task, instead of a more complex communication task, was in order to examine both word presentations and memory processes for dual-task effects. In this study, some of the secondary task word presentations were paired with drag force changes (analogous to breaking events requiring a reaction in Sawyer et al.,

2014), but there were also word presentations and force changes that occurred independently to one another. We expected dual-task reaction time impairments with reaction times to drag force changes being significantly slower when paired with word presentations than when the drag force changes are unpaired with word presentations. In addition, our design enabled us to look at the overall dual-task interference of the word recall paired with motor-tracking (not just looking at response times to drag force changes, but also overall tracking error). These results can then be compared to previous studies which have also employed the word recall task but paired with different tasks, such as climbing and running (Darling and Helton, 2014; Epling et al., 2016; Epling et al., 2018; Epling et al., 2018; Green et al., 2014; Green and Helton, 2011; Woodham et al., 2016).

In regards to the presence of an audio warning signal preceding a word presentation, participants need to devote more of their visual perception resources to scanning between the HMD and the tracking task. However, given the HMD is in the field of view, similar to Woodham et al. (2016) we expected that the presence of this warning signal would have little to no effect on overall dual-task performance. Previous studies have utilized audial warnings with HMD presentations paired with driving, but have not explored the impact of audial warnings per se (He et al., 2018). Our expectation, in line with previous research suggesting that the dual task interference between receiving messages and manual position control is largely due to cognitive, not perceptual, interference (He et al., 2014; Ishigami and Klein, 2009), was that audial warnings would have minimal impact on tracking performance.

## 1. Methods

### 1.1. Participants

Participants were 19 (14 men; 5 women) university students. An unexpected hardware fault, however, caused one of the female participants' data to be only partially recorded and her data were removed. The remaining 18 participants had a mean age of 23.12 years (SD = 2.91).

### 1.2. Design

The experiment was a within-participants design where each participant completed all five conditions; (1) a tracking alone condition, (2) a word memory alone condition, (3) a tracking and word memory condition, (4) a word memory preceded by audial signals condition, and (5) a tracking and word memory preceded by audial signals condition. An extended Latin square design was used so that each condition was preceded and followed by each other condition once in each position (see Fig. 1). This design has 20 possible row/combinations (see Fig. 1) so one was chosen at random to be removed to match the original sample size of 19. The primary performance metrics were words recalled and tracking performance.

### 1.3. Stimuli and apparatus

#### 1.3.1. Word task

The word memory task was presented on a Google Glass optical head-mounted display, with a 640 by 360 pixel resolution and 14° field of view. The device was resting within 5 cm of the participant's eye, elevated between 25 and 40° above the horizontal medial plane of the participant's visual field. The variance in elevation was due to the differences in participants' facial structure which was not individually measured. Four lists of 20 words (see Appendix A), one list for each condition utilizing the word recall task, were created using the online Paivio et al. Word List Generator (Paivio et al. 1968). Responses to this task were made verbally and scored by the experimenter. Input to the Glass, where necessary, was made using the touch pad on the side of the display. The Glass was synchronised with the desktop machine via a

|   |   |   |   |   |  |   |   |   |   |   |
|---|---|---|---|---|--|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 |  | 1 | 3 | 5 | 2 | 4 |
| 5 | 1 | 2 | 3 | 4 |  | 4 | 1 | 3 | 5 | 2 |
| 4 | 5 | 1 | 2 | 3 |  | 2 | 4 | 1 | 3 | 5 |
| 3 | 4 | 5 | 1 | 2 |  | 5 | 2 | 4 | 1 | 3 |
| 2 | 3 | 4 | 5 | 1 |  | 3 | 5 | 2 | 4 | 1 |
| 1 | 5 | 4 | 3 | 2 |  | 1 | 4 | 2 | 5 | 3 |
| 2 | 1 | 5 | 4 | 3 |  | 3 | 1 | 4 | 2 | 5 |
| 3 | 2 | 1 | 5 | 4 |  | 5 | 3 | 1 | 4 | 2 |
| 4 | 3 | 2 | 1 | 5 |  | 2 | 5 | 3 | 1 | 4 |
| 5 | 4 | 3 | 2 | 1 |  | 4 | 2 | 5 | 3 | 1 |

Fig. 1. The extended Latin square design.

Bluetooth connection and the task was run using custom code created in the Processing language (Processing.org, 2015) with the native Android bluetooth library.

### 1.3.2. Tracking task

The tracking task was presented on a 22-in. Phillips monitor (1680 × 1050 pixels). All stimuli for this task appeared within a 1024 × 768 pixel window at a viewing distance of approximately 70 cm. In the center of the window was a target comprised of five red concentric circles of radius 20, 40, 60, 80 and 100 pixels on top of a light grey background. The target was a white circle with a radius of 16 pixels with a 4 pixel black border. Fig. 2 shows the experiment screen layout. Participants responded to the tracking task with a Saitek ST90 joystick. Conditions were started using the joystick trigger and during the tasks all input was made using the joystick's tilt. Task presentation and data recording were run on an Intel Core i5 processor with a Windows 7 enterprise platform using custom code created in the Processing language (Processing.org, 2015) with the BluetoothDesktop library (Extrapixel.ch, 2015).

The tracking task program ran at 30 frames per second and on each frame, two separate motions were calculated. The first was the cursor drag motion which pulled the cursor in a random direction at a maximum speed of 1.5 pixels per frame (45 pixels/s; 12.7 mm/s; 2.5% screen width/s). Each drag motion lasted between one and 9 s with both a half-second fade in after a drag force change and half-second fade out time before the next drag force change. During these fade in periods the drag motion accelerated at a uniform rate from zero pixels per second to

the maximum speed and this was reversed for the fade out period. Once a drag motion had been completed, a new drag motion was chosen by selecting a random point from within the area of the program window and calculating the time it would take to travel to that point at the maximum speed, adjusting for the fades.

The second motion was the input from the participant via the joystick. The Saitek ST90 joystick's rectangular range of motion was mapped to a circular space to make the maximum velocity 5 pixels per frame (150 pixels/s; 42.3 mm/s; 9% screen width/s) in any direction. Measurements of the velocity of both motions (in pixels per frame), the magnitude of the composite vector created by the combination of the two motions (in pixels per frame) and the displacement of the cursor from the center of the target (in pixels) was sampled and recorded on every 6th frame (200 ms).

### 1.4. Procedure

Prior to the experiment, each participant came in for two practice sessions spaced a week apart. In each of these practice sessions the participants performed the compensatory tracking task for 8 min in order to reduce any learning effects occurring for the tracking task during the main experiment. Participants quickly reached a performance asymptote and did not improve across practice sessions. Word memory is an ability practiced in normal day to day life so there were no training sessions for the secondary task.

On arrival at the office, participants were given an explanation of the experiment, its purpose and the approximate amount of time required, after which consent to their participation was gained. Participants were informed for the word recall tasks they would have to recall the words at the end of the task. In the tracking task conditions participants were instructed to maintain the position of the cursor in the middle of the circle. The participant was then given an opportunity to familiarize themselves with Google Glass, adjust to the weight and sensation of wearing Glass and adjust the position and focus of the display prism so they could clearly see displayed words and numbers on the screen. A short navigation task on Glass familiarised each participant with the audio cues associated with Glass. When participants were comfortable with the operation, the experiment proceeded to the first condition.

Each participant was expected to complete all five experimental conditions: (1) a tracking alone condition, (2) a word memory alone condition, (3) a tracking and word memory condition, (4) a word memory preceded by audial signals condition, and (5) a tracking and word memory preceded by audial signals condition. Instructions were repeated before each task verbally and on the computer screen while the desktop and glass synchronised. Each condition ran for 4.5 min. In conditions without the tracking task, the computer screen showed only the background neutral grey colour with no cursor or target.

In the word recall task conditions, word presentations were displayed in the center of the Glass' prism in size 80 Georgia typeface. The text was white on a transparent background. The task began after a 15 s pause in all conditions. The purpose of this pause was to allow the participant to stabilize in the tracking task before presentations began. After this pause, the next 4 min were split into 20 blocks, each 8 s long with a 4 s spacing between them. Within each of these blocks, one word was presented at a random time drawn from a uniform distribution. Word presentations lasted 60 frames (2 s) and, in the conditions with audial warning signals, were preceded by a monotone beep which started 30 frames (1 s) before the presentation and lasted 15 frames (0.5 s). Participants were instructed to read the words aloud. This was recorded by the experimenter and used as a check to see if the participants were actually reading the words.

In all conditions, each word presentation had a 50% chance of triggering a drag force change for the tracking task's drag motion based on a pseudorandom binary string. These drag force changes had a minimum duration of 6 s, but otherwise shared all the same properties as a regular drag motion drag force change. For consistency across conditions, the

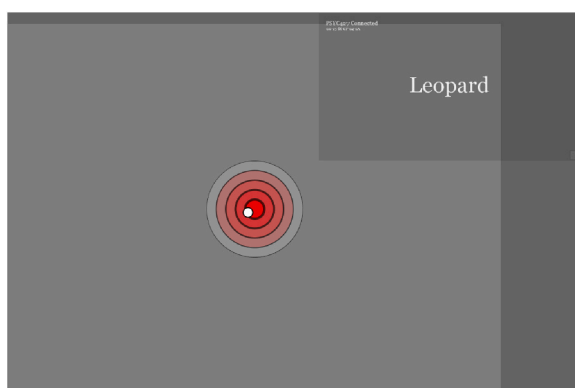


Fig. 2. The Google Glass image layered on top of the tracking task.

tracking task alone condition simulated these drag force changes using the same hidden background timer. There were approximately 10 paired drag force changes in each tracking task, where the drag force change was triggered by the word (in the tracking only condition no word was actually displayed). There were also approximately 16.7 drag force changes per tracking task that were unpaired with word presentations. The Google Glass was worn during the tracking alone condition but no images were displayed on the screen.

After all 20 word presentation blocks had passed, another 15 s pause was given, and then the task ended. The participant was notified of the end of each condition by a message on the desktop computer screen. At this time, the participants were asked to write as many of the words as they could remember on a response form. Participants were given 4 min for responses to each condition, but the full time was rarely used with the majority of participants noting that they could not remember any further words. A minute rest period was given between conditions.

## 2. Results

### 2.1. Word task

Across all conditions for all participants all 20 words were read aloud correctly, except in the tracking word presentation dual-task without a warning signal in which a single participant missed reading aloud 3 of the words. Any differences in word recall across conditions were therefore not due to a substantial failure in seeing or reading the words. We analysed the number of words recalled utilizing a 2 (single task – word recall alone – versus dual task – word recall with tracking) by 2 (warning signal absent versus warning signal present) repeated measures analysis of variance. Participants remembered significantly more words in the single word recall tasks ( $M = 11.94$ ,  $SD = 4.33$ ) than the word recall with tracking tasks ( $M = 10.44$ ,  $SD = 3.80$ ),  $F(1,17) = 4.92$ ,  $p = .040$ ,  $\eta_p^2 = 0.224$ ,  $M_{\text{difference}} = 1.50$ , 95% CI [0.07, 2.93], Cohen's  $d = 0.52$ . The absence of a warning signal ( $M = 10.97$ ,  $SD = 4.74$ ) versus the presence of a warning signal ( $M = 11.42$ ,  $SD = 3.20$ ) had no significant effect,  $F(1,17) = 0.40$ ,  $p = .538$ ,  $\eta_p^2 = 0.023$ , nor was there a significant interaction between the two factors,  $F(1,17) = 0.17$ ,  $p = .688$ ,  $\eta_p^2 = 0.010$ . Table 1 displays the mean number of words recalled for each condition.

### 2.2. Tracking performance

For each participant for each tracking task we calculated the mean square error (MSE) tracking displacement in pixels (see Table 1 for the mean MSE for each condition) We utilized two pre-planned orthogonal statistical contrasts for testing differences in mean square tracking error (Hays, 1973; Schad et al., 2020). The first test was between the tracking single-task with the average of the two tracking with word recall tasks, both no warning signal and warning signal conditions combined. Participants had significantly less tracking error in the tracking alone condition ( $M = 272.5$ ,  $SD = 166.1$ ) than in the tracking dual-task conditions ( $M = 381.0$ ,  $SD = 196.6$ ),  $t(17) = 4.39$ ,  $p < .001$ ,  $M_{\text{difference}} = 108.5$ , 95%CI [56.3, 160.7], Cohen's  $d = 1.03$ . For the second test, there was no significant difference between dual-task tracking error without a

warning signal ( $M = 374.8$ ,  $SD = 224.4$ ) and dual-task tracking error with a warning signal ( $M = 387.3$ ,  $SD = 204.0$ ),  $t(17) = 0.31$ ,  $p = .761$ ,  $M_{\text{difference}} = 12.5$ , 95%CI [-72.7, 97.7], Cohen's  $d = 0.07$ .

### 2.3. Post-event tracking performance: Finer grain analysis

Three types of events occurred during the tracking conditions in the experiment: word presentations on the HMD, drag force changes of drag which was being applied to the cursor, and paired word presentations on the HMD with drag force changes. Because the paired presentations created drag force changes under slightly different rules than the regular drag force changes, artificial “word” presentation times were generated for the tracking alone condition even though the HMD actually remained blank. Each participant had between nine and eleven word presentations, twenty four and thirty one drag force changes, and nine and eleven paired presentations. Figs. 3–5 display the average post-event effects on participants' tracking.

We calculated average reaction times by fitting a quadratic curve to the five points around the local maximum of the first post-event peak to estimate reactions times between the 200 ms measurement intervals. In cases where there were two equal local maxima, the earlier of the two was chosen as the center. Reaction time was then measured from event onset to this first peak.

For the word presentations events (see Fig. 3), as expected, since the participants in that task did not actually see any words, there were too few discernible post-event peaks in the tracking only task to be of comparative interest. As shown in Fig. 3, participants in the tracking only condition actually moved closer on average to the center relative to the initial position when the event started. We, therefore, excluded the tracking only condition from this analysis and only compared the two dual-task conditions. Participants in the tracking task without an auditory warning were significantly slower to react ( $M = 1407\text{msec}$ ,  $SD = 432$ ) than those with an auditory warning ( $M = 1090\text{msec}$ ,  $SD = 238$ ),  $t(12) = 2.57$ ,  $p = .024$ ,  $M_{\text{difference}} = 317\text{msec}$ , 95%CI [49, 585], Cohen's  $d = 0.71$ . The degrees of freedom being lower for this contrast was due to the inability to detect a discernible peak from some participants in one of the two dual-tracking tasks.

For the direction change events (see Fig. 4), similar to the overall tracking error analysis (see previous section), we performed two orthogonal statistical contrasts for testing differences in reaction times. There was no statistically significant difference between the tracking only task ( $M = 1293\text{msec}$ ,  $SD = 155$ ) and the dual-tasks combined ( $M = 1324\text{ msec}$ ,  $SD = 181$ ),  $t(17) = 1.09$ ,  $p = .290$ ,  $M_{\text{difference}} = 30\text{msec}$ , 95% CI [-28, 89], Cohen's  $d = 0.26$ . There was also no statistically significant difference between the tracking dual-task with an auditory warning ( $M = 1308\text{msec}$ ,  $SD = 183$ ) or without an auditory warning ( $M = 1339\text{msec}$ ,  $SD = 217$ ),  $t(17) = 0.79$ ,  $p = .443$ ,  $M_{\text{difference}} = 32\text{msec}$ , 95% CI [-54, 117], Cohen's  $d = 0.19$ .

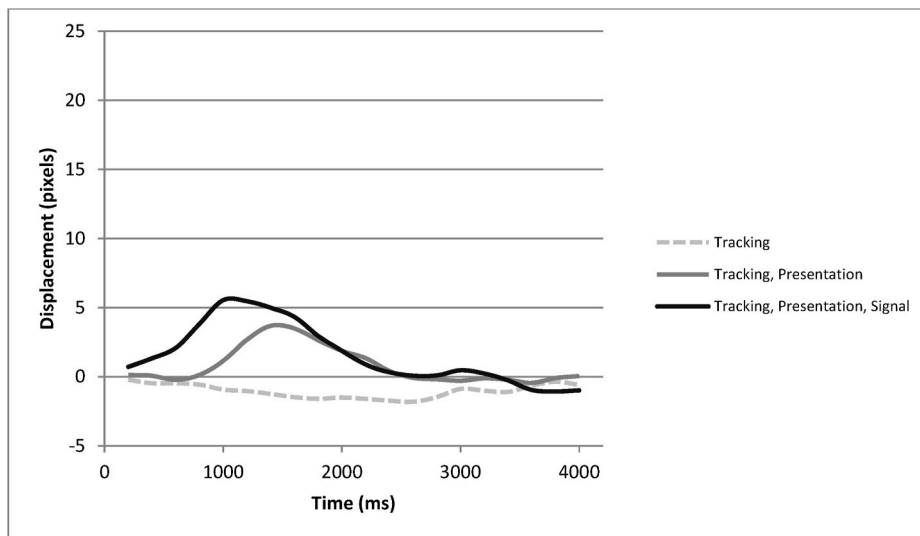
For paired word presentations with drag force changes (see Fig. 5), participants in the tracking only task were significantly faster to react ( $M = 1317\text{ msec}$ ,  $SD = 119$ ) than those in the dual-task tasks ( $M = 1528\text{msec}$ ,  $SD = 203$ ),  $t(17) = 5.21$ ,  $p < .001$ ,  $M_{\text{difference}} = 211\text{msec}$ , 95%CI [126, 297], Cohen's  $d = 1.23$ . Participants in the tracking task without an auditory warning were significantly slower to react ( $M = 1585\text{msec}$ ,  $SD = 208$ ) than those with an auditory warning ( $M = 1471\text{msec}$ ,  $SD = 235$ ),  $t(12) = 2.72$ ,  $p = .015$ ,  $M_{\text{difference}} = 114\text{msec}$ , 95%CI [26, 202], Cohen's  $d = 0.64$ .

## 3. Discussion

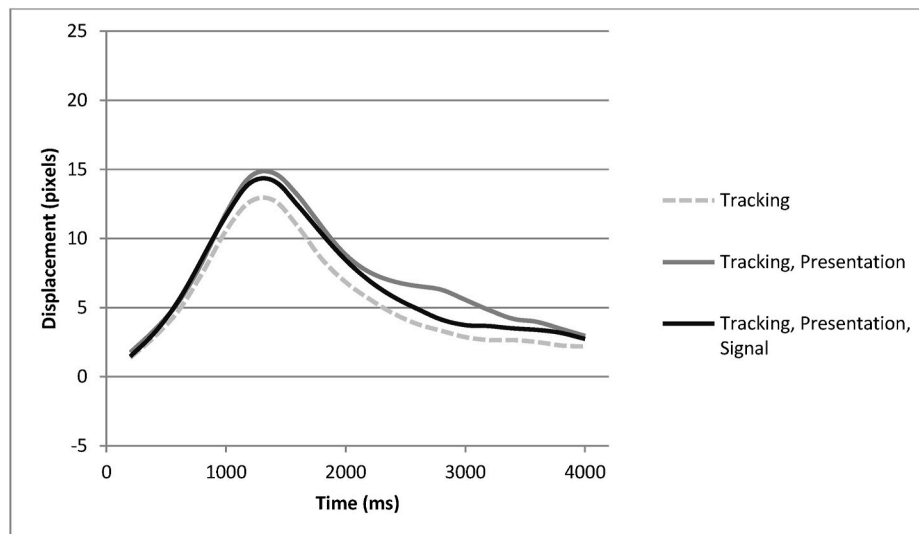
When compared to the findings of Woodham et al. (2016) in which they employed a similar experimental design but with free climbing instead of tracking, in both cases there was statically significant dual-task interference and in both cases the presence or absence of an auditory warning signal made no significant difference to overall task performance. There were, however, differences in the relative effects. In

**Table 1**  
Words recalled and tracking error for all conditions.

| Condition                      | Words Recalled |     | Tracking (MSE) |       |
|--------------------------------|----------------|-----|----------------|-------|
|                                | M              | SD  | M              | SD    |
| Presentation                   | 11.8           | 4.7 |                |       |
| Presentation, Signal           | 12.1           | 4.3 |                |       |
| Tracking                       |                |     | 272.5          | 166.1 |
| Tracking, Presentation         | 10.1           | 5.2 | 374.8          | 224.4 |
| Tracking, Presentation, Signal | 10.8           | 3.4 | 387.3          | 204.0 |



**Fig. 3.** Effect of word presentations on displacement averaged across participants. Displacement (y-axis) is measured from the pre-event (word presentation) value and is the distance between the cursor and target center (measured in pixels). Positive values in displacement indicate the cursor is moving further from the center than the cursor was when the event started; negative values in displacement indicate the cursor is moving closer to the center than the cursor was when the event started. Time (x-axis) is measured from the word presentation (ms). No word was actually presented on the HMD in the tracking alone condition.



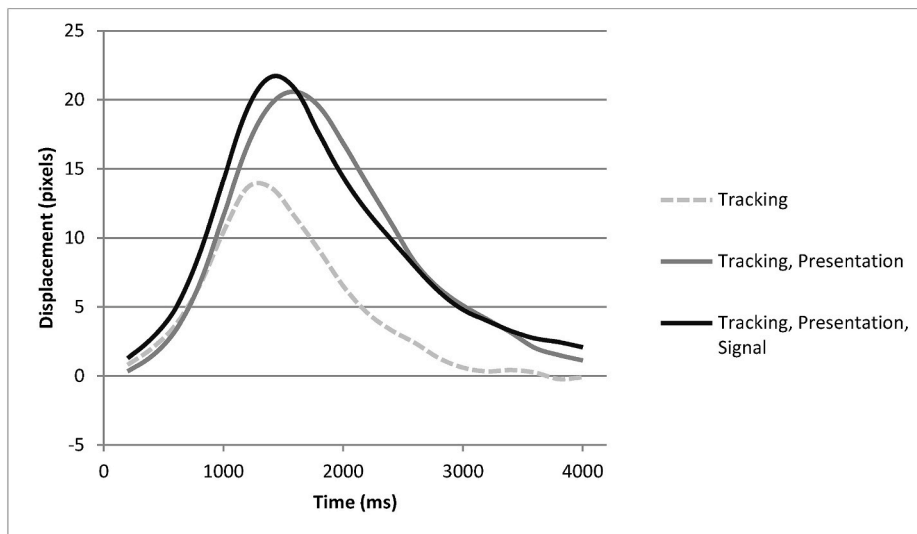
**Fig. 4.** Effect of drag force changes on displacement averaged across participants. Displacement (y-axis) is measured from the pre-event (drag force change) value and is the distance between the cursor and target center (measured in pixels). Positive values in displacement indicate the cursor is moving further from the center than the cursor was when the event started. Time (x-axis) is measured from the drag force change event (ms).

both experiments participants were not given task priority instructions. However, in the case of climbing, there was a much larger impact of climbing on later word recall (Cohen's  $d = 2.06$ ) than in the present case with tracking (Cohen's  $d = 0.52$ ). When comparing the loss of words recalled when dual-tasking versus when not, in the present study there was a 13% loss and in Woodham et al. (2016) there was a 45% loss. Climbing appears to have much larger impact on later word recall than seated manual tracking. In the present study, when comparing tracking error in the single-task and the dual-task we found a Cohen's  $d = 1.03$ , whereas for climbing, Woodham et al. (2016) when comparing climbing distance in a single-task condition and a dual-task condition found a Cohen's  $d = 0.64$ . The need to recall words had more impact on tracking in the present study than Woodham and colleagues found on climbing. This differential effect may be due to climbing taking natural priority due to the dangers of falling. Shumway-Cook and Woollacott (2000) proposed the "posture first hypothesis," in which people inherently give tasks that risk postural instability priority. This finding also matches a growing body of work using the dual-task method with posture risky movement tasks (Blakely et al., 2016; Blakely et al., 2021; Darling and

Helton, 2014; Epling et al., 2016; Epling et al., 2018; Epling et al., 2018; Green et al., 2014; Green and Helton, 2011; Woodham et al., 2016).

At a more fine grain level, when the drag force changes were paired with a word presentation on the HMD, reactions times increased by an average of 0.21 s (15%), an amount of time similar to that found for both handheld and hands-free mobile phones in other experiments (Caird et al., 2008). This means, despite the user's ability to see through the HMD's screen, they still were unable to attend to both tasks at the same time. In the simulated driving experiment by Sawyer et al. (2014) where they also employed a HMD, the average response time between the lead car braking and the depression of the brake increased from 1.3 to 1.7 s, twice the increase found in this study.

This is likely due to the timing differences in the workload imposed by the secondary tasks used in the present study (word presentation with later recall) and by Sawyer et al. (arithmetic). In arithmetic tasks, cognitive workload as measured by general arousal via pupil dilation increases sharply during the calculation period, whereas in memory tasks, cognitive workload only increases slightly as each new item is added to the participants phonological loop (Klingner et al., 2010).



**Fig. 5.** Effect of paired word presentations and drag force changes on displacement averaged across participant. Displacement (y-axis) is measured from the pre-event (drag force change paired with a word presentation) value and is the distance between the cursor and target center (measured in pixels). Positive values in displacement indicate the cursor is moving further from the center than the cursor was when the event started; negative values in displacement indicate the cursor is moving closer to the center than the cursor was when the event started. Time (x-axis) is measured from the drag force change event (ms). A drag force change alone was made in the tracking alone condition (there was no word actually presented in this condition on the HMD).

The presence of the auditory warning signal caused participants to respond faster to paired presentations by 0.11 s, perhaps providing evidence for some form of perceptual interference mitigated by the auditory alert. However, despite users reacting faster to the paired drag force change, the amount of time it took to compensate for the drag force change did not appear to decrease and the initial displacement from center was actually slightly greater for the audial warned paired presentations (see Fig. 5). Indeed, while there was no statistically significant difference between the dual-tasks for overall tracking task mean square error (MSE), the audial warned dual-task actually had a slightly higher average MSE than the non-warned dual-task. One explanation for this pattern of results is that the presence of the signal only shifted the order in which the participants attended to the word and responded to the drag force change rather than improving how the components were performed concurrently. Current design standards for hand-held devices make liberal use of auditory signals (Developer.android.com, 2014; Developer.apple.com, 2014), but with HMDs always being within the user's field of view, these signals may be unnecessary and, at worst, interfere with situation appropriate sequencing of attention. Future research should investigate the perceptual costs of scanning an overlaid visual channel at different focal lengths as well as if these costs are moderated by visual clutter. The primary tracking task in this study had minimal visual clutter which will usually be present in a real-world tasks and is known to make visual judgements more difficult (Steelman et al., 2013). In more visual cluttered situations, perhaps auditory prompts would be found more effective than found here or in Woodham et al. (2016).

The presence of both post-event and sustained interference while dual tasking with the HMD matches the findings of both simulated and real-world studies of dual-task mobile phone use while driving (Caird et al., 2008; Consiglio et al., 2003; Horrey and Wickens, 2004; Hosking et al., 2009; Strayer et al., 2006) as well as the early studies on dual-task HMD use (He et al., 2015, 2018; Tippey et al., 2017; Sawyer et al., 2014; Young et al., 2016).

While the present study employed an artificial word recall task, our findings are in line with studies employing more natural communication tasks. The compensatory tracking task utilized in this study was also different from previous HMD studies employing driving simulators or natural movement tasks (e.g., climbing). Compensatory tracking with a joystick control does, however, emulate some occupational tasks. For example, many agricultural, construction, forestry, and mining machines with a loader, such as excavators, forklifts, skid-steer loaders, tractors, tele-handlers, and wheel loaders, utilize joystick type controls to control the loader. Forces can perturb the loader during operations

and these need to be compensated for by the operator or there is a risk of losing a load, damaging nearby structures, or worse, tipping the machine over. While the compensatory tracking task here is far from a high fidelity machine simulator, there are some fundamental similarities in task demands. Given the increasing availability and interest in employing HMDs in these operations (Sitompul and Wallmyr, 2019) research examining the use of HMDs during manual tracking tasks with ever increasing fidelity is warranted. Higher fidelity tasks are needed well beyond the task employed in this study, as operators do not just control the movement of the loader alone, but also the machine itself, and often in cluttered and complex three dimensional environments.

In conclusion, using HMDs causes users to react more slowly to changes as well as impairing their ability to maintain a position. The present study's results when compared to those of Woodham et al. (2016) also suggest people are likely to attempt to ignore or deprioritize the workload demands imposed on them by the HMD in line with the Shumway-Cook and Woollacott (2000) posture-first hypothesis during tasks where postural stability is at risk, such as climbing. A future research question is whether this posture-first hypothesis would extend to tasks beyond natural postural tasks, such as standing, walking, running, and climbing, to operations where an operator is liable to tip-over a machine they are operating (Rondelli et al., 2018). This issue of task prioritization while dual-tasking with a HMD should be investigated in future studies, as in many applied settings the operator can freely choose their own task prioritization.

#### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

#### Appendix A

Word List 1: "slipper", "glacier", "pepper", "settler", "odour", "dreamer", "cigar", "captive", "banner", "poster", "arrow", "piston", "pencil", "hardwood", "beggar", "invoice", "robber", "elbow", "butcher", "salute".

Word List 2: "candy", "reptile", "comrade", "sunset", "shotgun", "profile", "scarlet", "harness", "bandit", "portrait", "barrel", "sunburn", "warbler", "lobster", "rubble", "bouquet", "hostage", "cellar", "tweezers", "abyss".

Word List 3: "fabric", "mucus", "hurdle", "leopard", "pianist", "footwear", "cuisine", "hammer", "trellis", "garments", "nectar",

“sultan”, “trumpet”, “doorman”, “hillside”, “pudding”, “kettle”, “daybreak”, “speaker”, “portal”.

Word list 4: “costume”, “saloon”, “kerchief”, “python”, “fiord”, “bullet”, “mammal”, “salad”, “lemon”, “twilight”, “abode”, “jelly”, “singer”, “goblet”, “painter”, “wigwam”, “typhoon”, “basement”, “sulphur”, “juggler”.

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